

Scarcity, Work & Choice

Unit 3 — Study Guide & Lecture Notes

- › Preferences, utility & indifference curves
- › The feasible frontier, MRS and MRT
- › The utility-maximising choice: $MRS = MRT$
- › Income & substitution effects and working hours

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 3 slides section-by-section and expands the reasoning so you can read it on its own or annotate it in class. Every diagram — indifference curves, the feasible frontier, the tangency condition, and the income/substitution decomposition — is redrawn cleanly, with the convex curves computed rather than sketched so the geometry is exactly right.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

The thread from Units 1–2

- **Unit 1:** labour is **work**.
- **Unit 2:** labour is an **input** in production, and new technologies raise the **productivity of labour**.
- **Unit 3 (this unit):** if technology makes an hour of work more productive (higher wage), **how does an individual respond?** Do they take the gain as more **consumption**, more **free time**, or both?

THE QUESTION THIS UNIT ANSWERS

Build a **model of individual choice** that explains the differences in **work hours** across countries and over time — and, in particular, why rising wages have historically been taken partly as **more leisure**.

Outline

1. **A.** Introduction — the puzzle of hours and income
2. **B.** Scarcity and choice: preferences, indifference curves, MRS, the feasible frontier and MRT
3. **C.** Decision-making under scarcity: the optimal choice, $MRS = MRT$
4. **D.** Income and substitution effects
5. **E.** Application to technological change

SECTION A

Introduction

Since 1870, living standards have risen enormously in most countries. But the gains have **not** all been taken as extra income: there are large disparities in **free time** and **income** across countries. Some rich countries work long hours; others have converted much of their productivity gains into shorter working weeks and longer holidays.

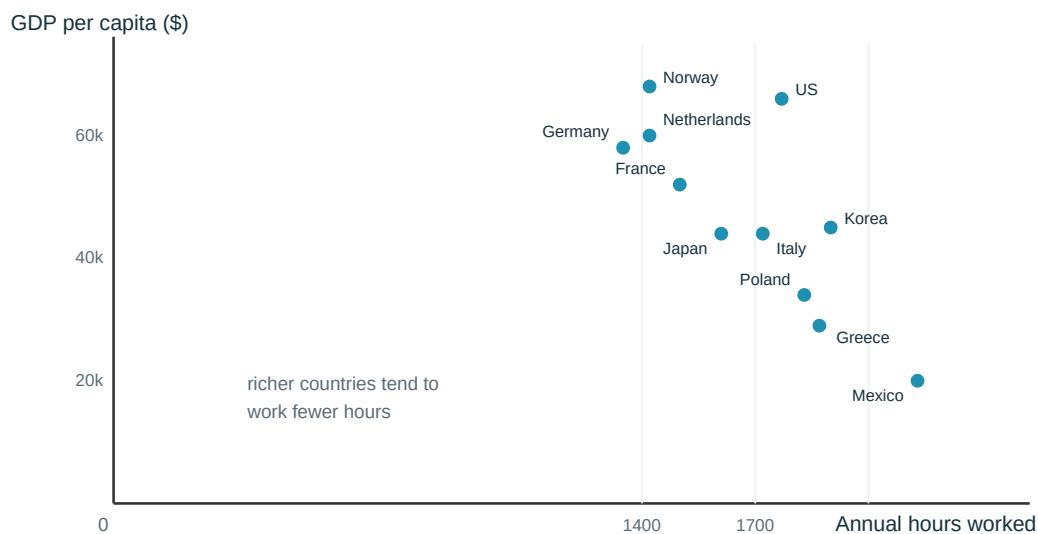


Figure 1. Across richer countries, higher income per person tends to come with **fewer** annual hours worked (e.g. Germany, the Netherlands, France) — while some equally rich countries (e.g. the US, Korea) work much more. Income alone does not pin down hours; **preferences** (and culture, and politics) do. This unit builds the model that makes sense of the scatter.

LISTEN FOR IN THE LECTURE

The framing question for the whole unit: *a new technology raises the productivity of labour* — *how does that affect (i) living standards and (ii) the free time and working hours people choose?* Keep both halves in mind; the answer turns out to hinge on two opposing forces (income and substitution effects).

SECTION B

Scarcity & Choice: Key Concepts

Individuals want to consume as much as possible **and** have as much free time as possible. But a day has only a limited number of hours — this is **scarcity**. Increasing consumption means **sacrificing free time** (you must work to earn). So we need a model of how a person picks the best combination of the two.

TWO INGREDIENTS OF THE MODEL OF CONSTRAINED CHOICE

- **Preferences** — how the individual ranks combinations of consumption and free time.
- **The trade-off (constraint)** — the feasible combinations of consumption and free time, i.e. how much consumption an hour of free time costs.

Simplifying assumptions

- Individuals **don't care about the future**, so there are **no savings**.
- Individuals **cannot borrow**.
- Therefore **income = consumption** (whatever you earn, you spend now).

WHY MAKE "UNREALISTIC" ASSUMPTIONS?

Recall from Unit 2: a model deliberately omits detail to expose the essential mechanism. Ruling out saving/borrowing lets us focus cleanly on the one trade-off that matters here — **free time vs consumption**.

Preferences and utility

People like **both** free time and consumption, and can **rank** the many possible combinations.

DEFINITION — UTILITY

Utility is a numerical indicator of the value a person places on an outcome. Higher-utility combinations are preferred to (chosen over) lower-utility ones.

Indifference curves

DEFINITION — INDIFFERENCE CURVE

An **indifference curve** shows all combinations of two goods that give the **same utility** (the person is "indifferent" between them).

Four properties you must know and be able to justify:

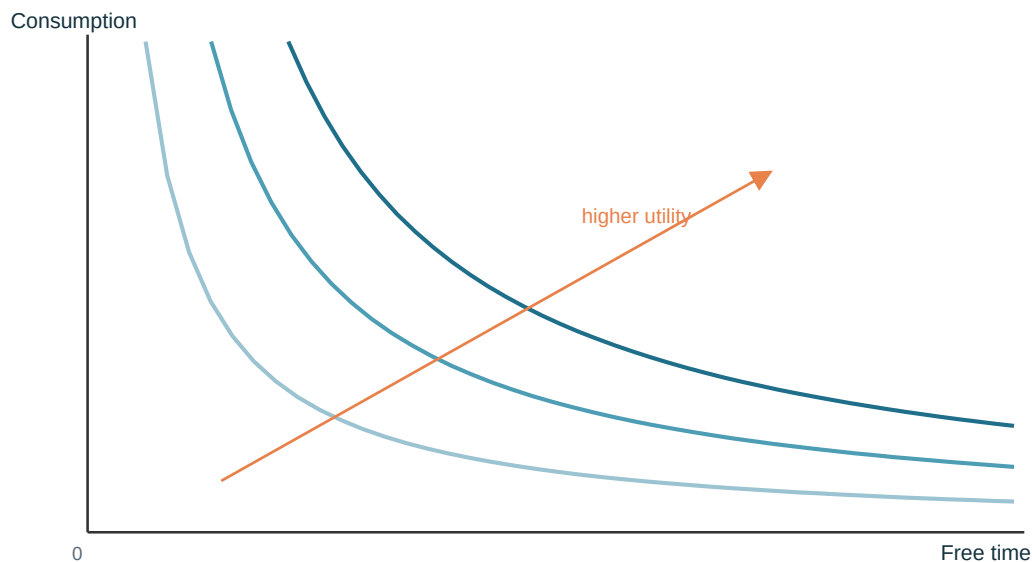


Figure 2. A family of indifference curves for free time and consumption. **Property 1:** they slope **downward** (more is better, so to stay equally happy you give up one good to get more of the other).

Property 2: curves **further from the origin** mean **higher utility** (more of both goods).

THE FOUR PROPERTIES

1. **Downward-sloping** — because more of a good is better.
2. **Higher curves = higher utility** — they contain bundles with more of both goods.
3. **They never cross** (see below).
4. **They get flatter to the right** — a diminishing marginal rate of substitution (convex to the origin).

Property 3 — why indifference curves cannot cross

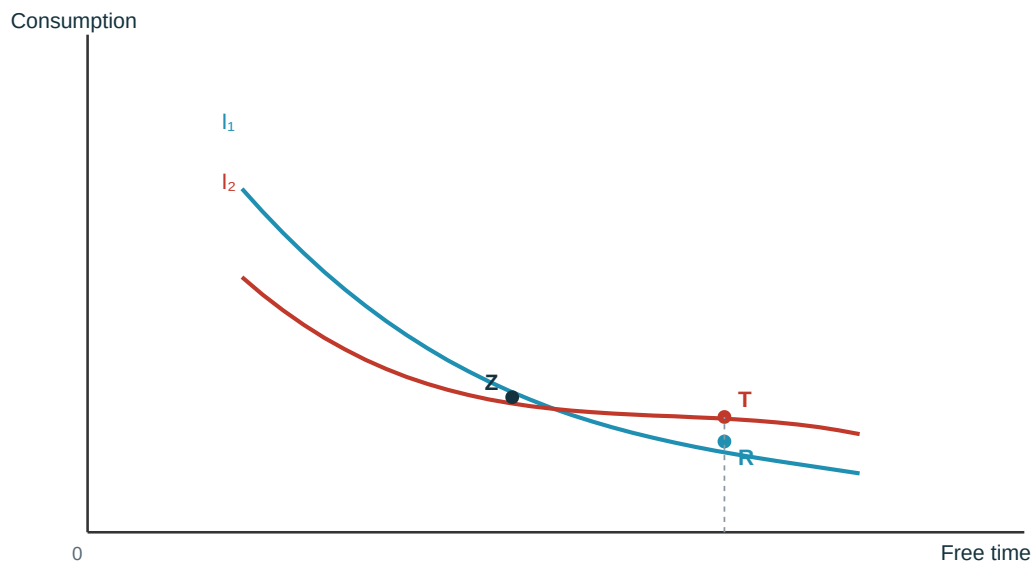


Figure 3. Suppose two indifference curves crossed at Z . Take points R (on I_1) and T (on I_2) directly above one another. Being on the same curve as Z , R gives the same utility as Z , and T also gives the same utility as Z — so R and T must be equally good. But T has **more consumption than R for the same free time**, so T is strictly better. Contradiction — therefore curves **cannot cross**.

Property 4 — the marginal rate of substitution (MRS)

DEFINITION — MARGINAL RATE OF SUBSTITUTION

The **MRS** is the amount of one good a person is **willing to give up** to get one more unit of the other, holding utility constant. Graphically it is the **(magnitude of the) slope of the indifference curve** at a point. It **diminishes** as you move right: the more you already have of a good, the less of the other you'll sacrifice for still more of it.

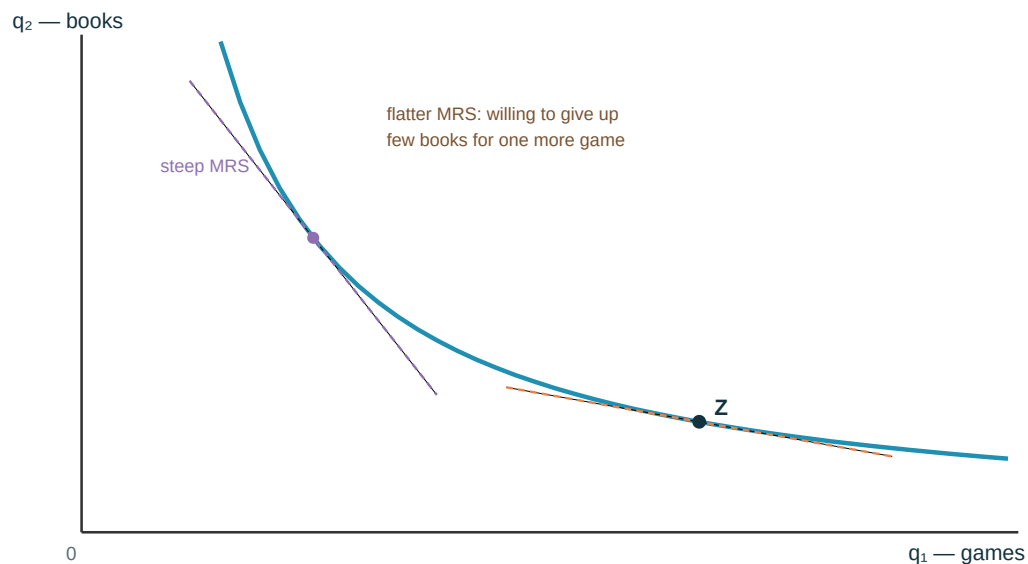


Figure 4. A consumer choosing between games (q_1) and books (q_2). At bundle **Z** she already has **many games and few books**, so she will give up only a **few books** for one more game — the curve is **flat** (small MRS). Far to the left, with few games, she'd sacrifice **many** books for a game (steep, large MRS). The MRS is the slope of the tangent.

LISTEN FOR IN THE LECTURE

The MRS is a **subjective willingness to trade**, read off preferences. Later it will be set equal to the MRT (an **objective** trade-off read off the constraint) to find the optimum. Keep the two straight: **MRS = preferences, MRT = constraint**.

Putting a Price on the Priceless

The MRS is powerful because it lets us value things that have **no market price** — marriage, children, health. There's no market, so no observed monetary value, and asking "how much would you pay for...?" directly is unreliable. The trick is **indirect measurement via happiness**.

- Ask people "**How satisfied are you with your life?**" and relate the answer to observable characteristics (income, marital status, health, employment...).
- The **MRS between, say, marriage and income** then reveals the willingness to pay for marriage — how much extra income would compensate a single person for being unmarried, at constant happiness.

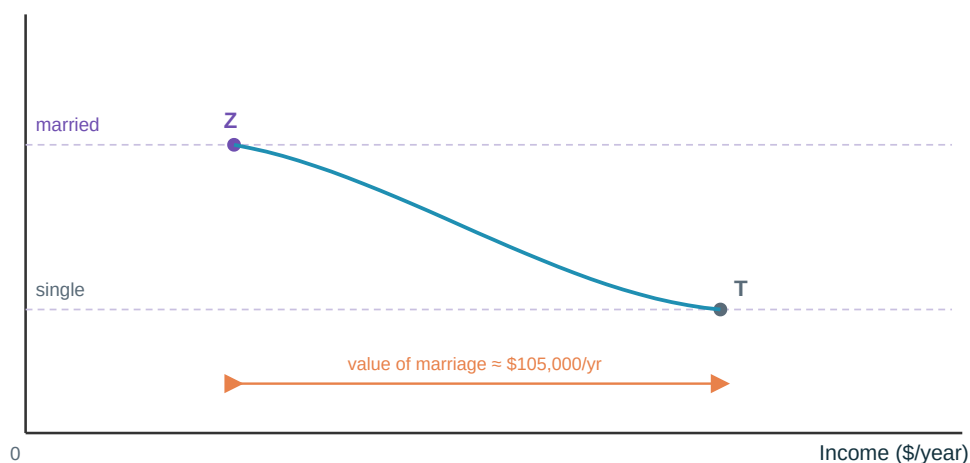


Figure 5. Same-happiness (indifference) reasoning applied to marriage. A married person at income Z is exactly as happy as a single person at the higher income T. The horizontal gap — the extra income needed to keep a single person as happy — is the **monetary value of marriage**, estimated at roughly **\$105,000 per year**.

What?	In comparison to?	Value (per year)
Excellent health	Very bad health	+\$463,170
Married	Single	+\$105,000
Divorce	No divorce	−\$34,000
Unemployed	Full-time job	−\$114,248
Death of partner	No death of partner	−\$308,780

Source: estimates via life-satisfaction regressions (Blanchflower & Oswald-style; slide cites bakadesuyo.com). Signs show whether the event raises or lowers well-being.

But do we get used to everything? Hedonic adaptation

A caveat to these numbers: over time people tend to **return to a long-run "set-point"** of happiness after major life events. The initial jump (or drop) in satisfaction fades.

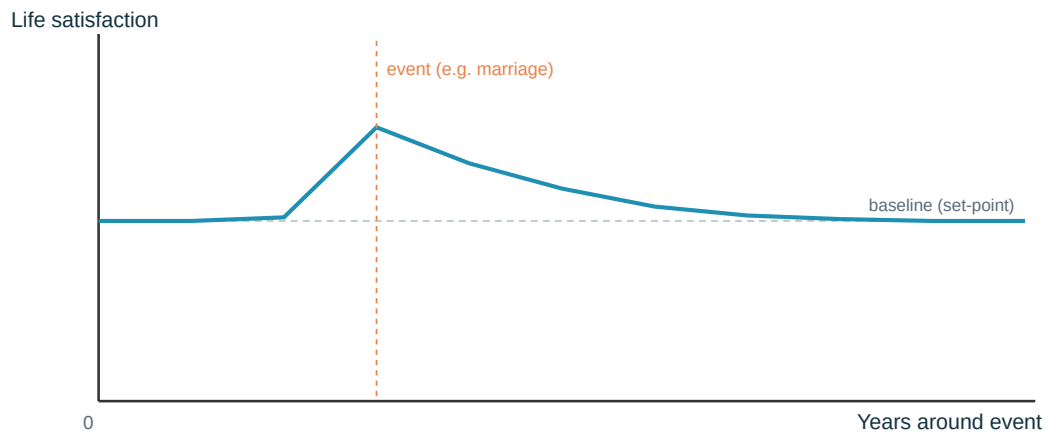


Figure 6. Life satisfaction around a major event (e.g. marriage). There is a spike, but within a few years people drift back toward their **baseline**. (Source: Clark, Diener, Georgellis & Lucas, 2008.) This **adaptation** means the money-values above should be read as short-to-medium-run, not permanent.

The Feasible Frontier & the MRT

Preferences tell us what the person **wants**; now we need what is **possible**. The trade-off between consumption and free time is drawn as the **feasible frontier**.

DEFINITIONS

- **Feasible frontier** — the combinations of free time and consumption that can just be **reached** (work all available hours → max consumption, no free time; work none → max free time, no consumption).
- **Feasible set** — the frontier **plus everything inside it** (all attainable combinations).

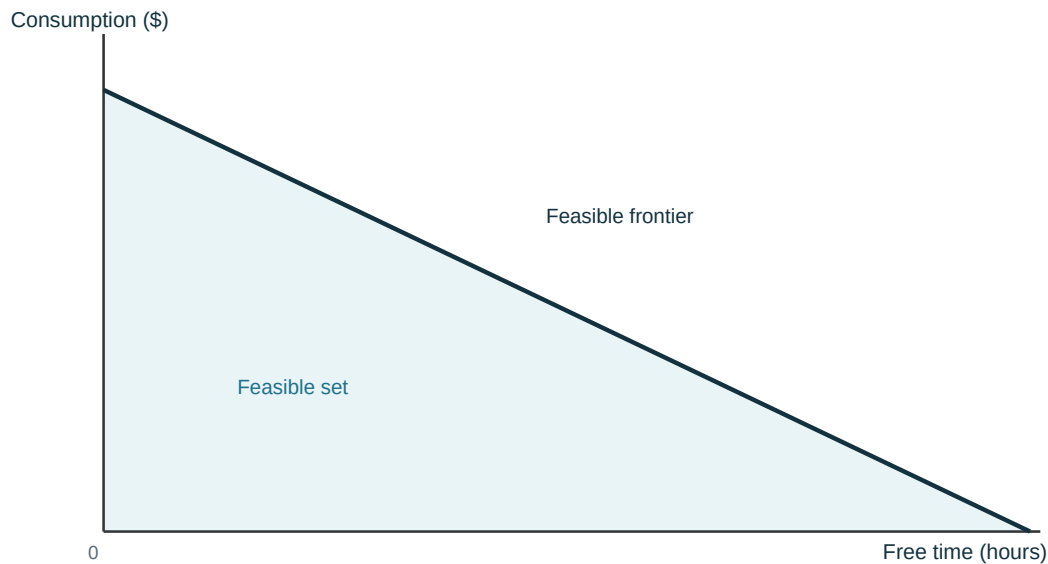


Figure 7. The feasible frontier and the feasible set. Points on the frontier use all available time; points inside waste some potential consumption or time; points outside are unattainable.

DEFINITION — MARGINAL RATE OF TRANSFORMATION

The **slope of the feasible frontier** is the **opportunity cost of free time** — the consumption you forgo for one more hour off. This is the **marginal rate of transformation (MRT)**: the rate at which free time can be **transformed** into consumption (i.e. the wage).

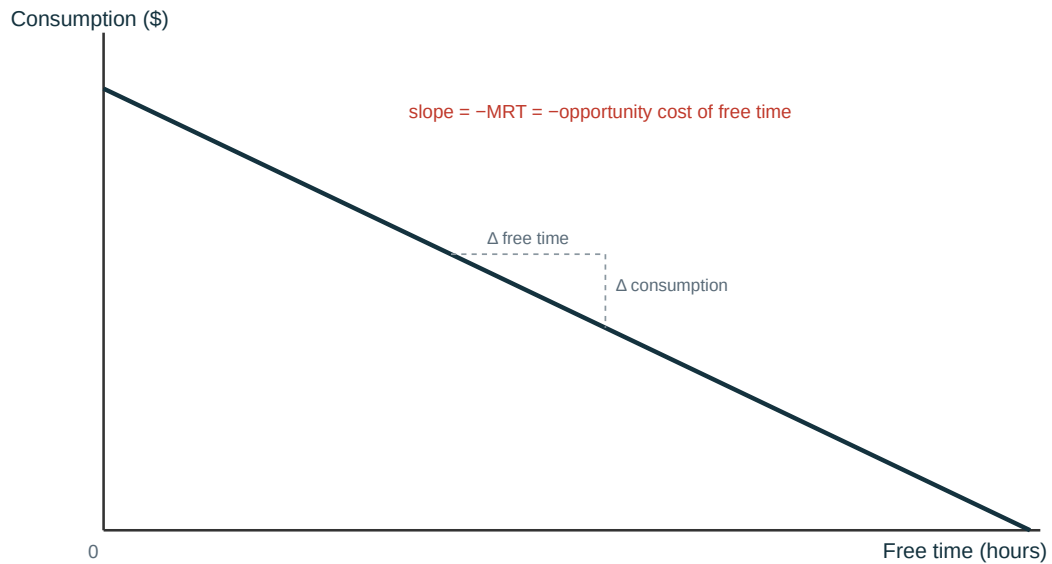


Figure 8. The MRT is the frontier's slope. If an hour of free time costs $\$w$ of consumption (the wage), the frontier has slope $-w$ and the $MRT = w$. A **higher wage** → **steeper frontier** → **higher opportunity cost of free time**.

CHECK YOURSELF

- Why must indifference curves be convex (flatten to the right)? What would a *concave* one imply about willingness to trade?
- The MRS and MRT are both "slopes" — whose slope is each, and what does each measure (willingness vs ability to trade)?

SECTION C

Decision-Making Under Scarcity

Now combine the two halves. Which feasible combination will the person choose? This is a **constrained choice problem**: pursue an **objective** subject to a **constraint**.

THE CONSTRAINED OPTIMISATION

- **Objective:** maximise **utility**.
- **Constraint:** the **feasible frontier**.

The optimal choice is the combination lying on an indifference curve **as far from the origin as possible** while still being feasible.

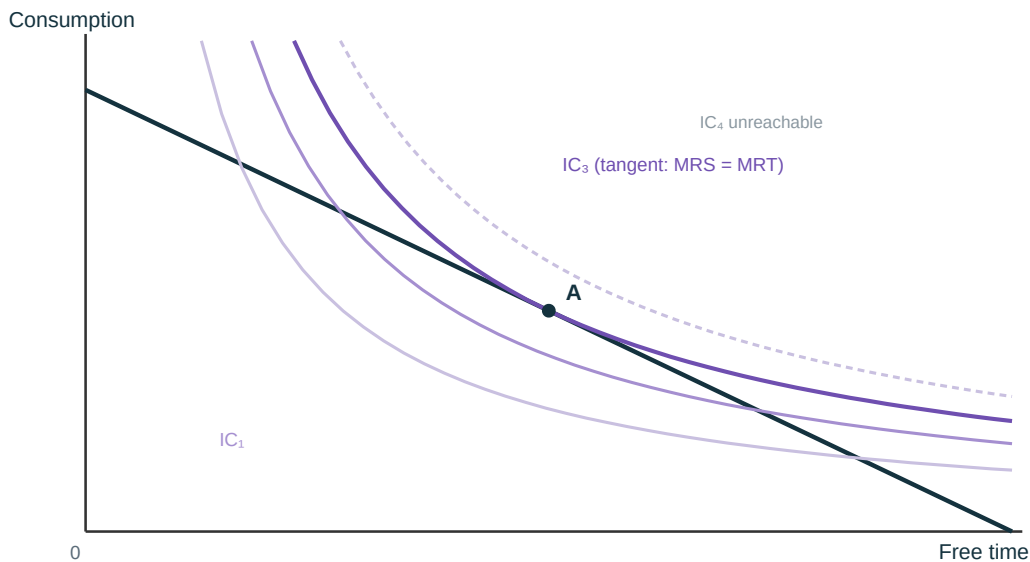


Figure 9. Overlay the indifference curves on the feasible frontier. IC_4 would be nicer but is **unreachable**; IC_1 is feasible but wasteful (you can do better). The best attainable curve just **touches** the frontier — the tangency point **A**.

OPTIMALITY CONDITION — $MRS = MRT$

The utility-maximising choice is where the **indifference curve is tangent to the feasible frontier**. At that point the rate at which the person is **willing** to trade the goods (MRS) equals the rate at which they **can** trade them (MRT):

$$MRS = MRT$$

Why the tangency is best — the intuition

At a non-tangency point the two rates differ, and you can always improve. Consider a point **D** where the indifference curve is **steeper** than the frontier ($MRS > MRT$):

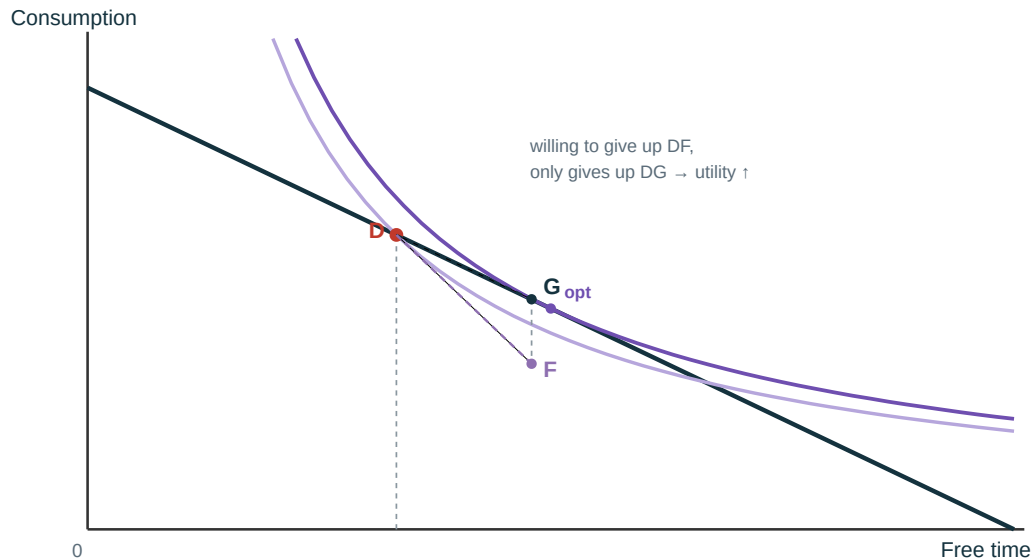


Figure 10. At D , to gain one more hour of free time the consumer is **willing** to give up DF of consumption (her steep MRS), but the frontier only **makes** her give up DG (the flatter MRT). Since she sacrifices **less than she was willing to** ($DG < DF$), taking more free time raises her utility — she moves to a higher indifference curve. She keeps going until $MRS = MRT$ (the optimum).

LISTEN FOR IN THE LECTURE

The logic is completely general: whenever $MRS \neq MRT$ there is an unexploited gain, so only the tangency can be optimal. If $MRS > MRT$, take more free time; if $MRS < MRT$, work more. This is the same "equate the margins" idea you'll meet again and again in microeconomics.

SECTION D

Income & Substitution Effects

Set-up — the summer-job example

You are planning your **10-week summer break**. You can work at **\$90 per day**, but you also value time for friends, holiday and study. How many days should you work?

Your feasible frontier has slope **-\$90/day** (the wage = the MRT = the opportunity cost of a day off). The optimum is the tangency **A**.

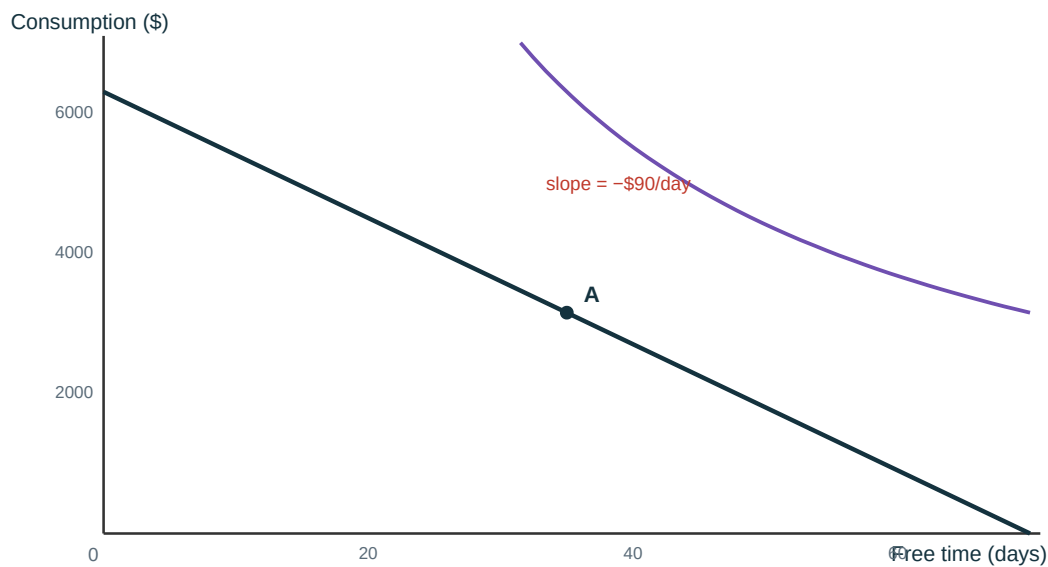


Figure 11. The summer-job feasible frontier: work all days → maximum consumption, no free time; take all days off → maximum free time, no consumption. Slope = $-\$90/\text{day}$. The chosen point A is where $\text{MRS} = \text{MRT}$.

A pure increase in income (a gift)

Your mother gives you **\$1000**, no strings attached. This does **not** change the wage, so the frontier shifts **up in parallel by \$1000** — the opportunity cost of free time is unchanged. You re-optimize at a new tangency.

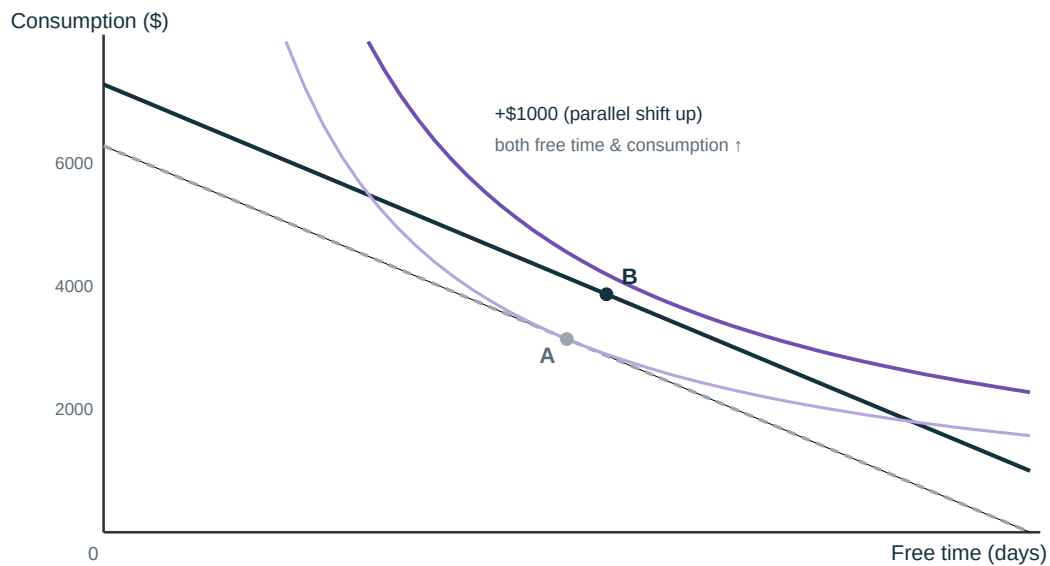


Figure 12. A \$1000 gift shifts the frontier up in **parallel** (same slope). The new optimum B typically has **both more consumption and more free time** — free time is a **normal good**. (With particular preferences free time might stay the same, but it should never **fall**.)

KEY TAKEAWAY

A rise in **unearned** income (no change in the wage) has only an **income effect**: more of everything you value, including free time. We expect free time to **rise or stay equal** — **never fall**.

A Wage Rise: Two Opposing Effects

Now a genuinely different job pays **\$130 per day**. Unlike the gift, a higher **wage** does two things at once, and they pull free time in **opposite** directions:

THE TWO EFFECTS OF A WAGE INCREASE

- **Income effect** — at your old hours you now earn more, so you can afford more of everything, including **more free time** (free time is normal). **Pushes free time up.**
- **Substitution effect** — free time is now more **expensive** (each hour off costs \$130 rather than \$90 of consumption), so you substitute away from it. **Pushes free time down.**

We separate them with a **hypothetical** frontier: a line with the **old slope** ($-\$90$) drawn tangent to the **new, higher** indifference curve.

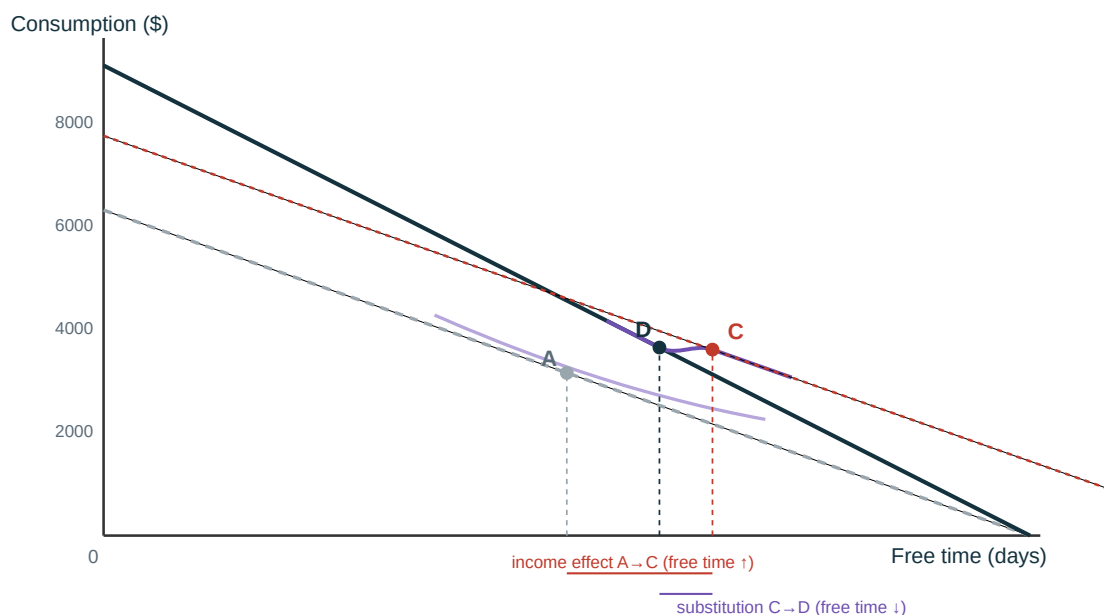


Figure 13. Decomposing a wage rise. **A → C (income effect):** a parallel move to the new higher indifference curve at the **old** relative price — free time rises. **C → D (substitution effect):** along the **same** indifference curve, the frontier steepens to the true new wage — free time falls. The **net** move A → D here shows free time **rising**, so the income effect dominates.

THE OVERALL EFFECT

$$\text{Overall effect} = \text{Income effect} + \text{Substitution effect}$$

Because the two effects have **opposite signs** for free time, the net direction depends on **which is larger** — and that depends on **individual preferences**. Free time (and hence hours worked) can rise **or** fall when the wage rises.

LISTEN FOR IN THE LECTURE

The construction is fiddly the first time. Anchor it on three points: **A** (old optimum), **D** (new optimum, on the true new frontier), and **C** (the hypothetical point — **same slope as old, same indifference curve as new**). Income effect = A → C, substitution = C → D. The consumption axis reveals the puzzle's resolution; the free-time axis reveals its sign.

Application to Technological Change

Differences over time

Now use the model to explain the historical fall in working hours. Take a US worker in **1900** at point A. Over the century, technological progress raised the wage, **steepening** the budget constraint. Decompose exactly as in Figure 13:

- **Income effect:** higher wage → afford more → **free time and consumption both rise** (A → C).
- **Substitution effect:** free time dearer → **free time falls**, consumption rises (C → D).
- **Net:** historically the **income effect dominated**, so **free time increased** — the working week and working year got shorter as countries got richer.

THIS IS EXACTLY FIGURE 13

The "technological change over time" diagram **is** the wage-rise decomposition, with the income effect winning. That is the model's explanation for the long-run decline in hours worked in rich economies.

Cross-country differences

Why do equally rich countries choose such different hours (Figure 1)? The model's baseline answer is **differences in preferences**. But preferences are shaped by more than taste:

- **Culture / norms** — differing attitudes to work and leisure.
- **Politics** — legal limits on working hours, mandated holidays.
- **Social preferences** — e.g. "keeping up with the Joneses": if status depends on relative consumption, people over-work.

Is this a good model?

WEAKNESSES

Not literally realistic: people don't compute MRS/MRT, and **most cannot freely choose their hours**.

WHY IT STILL WORKS

A good **approximation**: over time people **learn** the mix that suits them; hours shift through culture and politics (**indirect** choice); and people choose **which jobs to apply for**. It **explains real patterns** — hours across countries and over time — via preferences and income/substitution effects.

Summary, Formulas & Glossary

The unit in one paragraph

A person chooses between **consumption** and **free time** under scarcity. **Preferences** are drawn as convex, non-crossing **indifference curves**, whose slope is the **MRS** (willingness to trade). The **feasible frontier** is the constraint, whose slope is the **MRT** = the wage = the opportunity cost of free time. The best feasible bundle is the **tangency** where **MRS** = **MRT**. A pure income gain shifts the frontier out in parallel (only an **income effect**: free time rises or stays equal). A **wage rise** rotates the frontier and triggers **two opposing forces** — an income effect (free time $\uparrow$) and a substitution effect (free time $\downarrow$) — whose net depends on preferences. Historically the income effect has dominated, so richer societies work **fewer** hours.

Formula & concept sheet

Concept	Meaning / expression
Utility	numerical ranking of bundles; higher = preferred
Indifference curve	bundles with equal utility; convex, downward, never cross
MRS (marginal rate of substitution)	slope of indifference curve = willingness to trade
Feasible frontier	consumption = wage $\times$ (hours available - free time)
MRT (marginal rate of transformation)	slope of frontier = wage = opp. cost of free time
Optimal choice	MRS = MRT (tangency)
Income effect (wage $\uparrow$)	free time $\uparrow$ (normal good)
Substitution effect (wage $\uparrow$)	free time $\downarrow$ (leisure dearer)
Overall effect	income effect + substitution effect

Glossary

Term	Meaning
Scarcity	Limited time (and resources) forces trade-offs.
Constrained choice	Maximising an objective (utility) subject to a constraint (the frontier).
Utility	Numerical index of preference satisfaction.
Indifference curve	Equal-utility combinations of two goods.
MRS	Willingness to give up one good for one more unit of the other; slope of the IC; diminishing.
Feasible frontier / set	Attainable boundary / all attainable bundles.
MRT	Rate free time converts into consumption; slope of the frontier; the wage.
Income effect	Response to being richer at unchanged relative prices.
Substitution effect	Response to a changed relative price at constant utility.
Normal good	Demand rises with income (free time is one).
Hedonic adaptation	Tendency to return to a happiness set-point after events.

Exam-style questions to practise

1. Draw a feasible frontier for a wage of \$15/hour over a 16-hour day and mark the optimum. What is the MRT?
2. Explain, using MRS and MRT, why a point inside the feasible set can never be optimal.
3. A worker's wage doubles but she works *the same* hours. What does this tell you about the relative size of her income and substitution effects?
4. Use an income/substitution diagram to explain why average annual hours fell across the 20th century in rich countries.
5. Two countries have identical technology and wages but very different working hours. Give three model-consistent explanations.
6. Why can indifference curves not cross? Prove it with a diagram.

and tangencies are geometrically accurate; data figures are clean redrawings of the lecture charts for study purposes.